

Designing a Noise Reduction Filter for Voice Signals

From fixed low-pass filtering to Wiener-style frequency-domain masking

ECE0402 • Signals, Systems & Probability

Junze Xie, Ming Chen, Zifan Xu, Yimin Chen & Wanting Luo

Why this problem matters

Voice denoising as a real signal-processing task

Voice recordings and public announcements often lose clarity when background noise is present.

High-frequency hiss is a useful starting point because it is common, audible, and frequency-domain friendly.

Our goal is to connect a real application to Signals and Systems ideas through a controllable demo.

Real problem

Controllable model

Explainable system

Scenario and scope

Real-world motivation, controlled simulation

Scenario: a public-announcement style voice signal affected by high-frequency hiss.

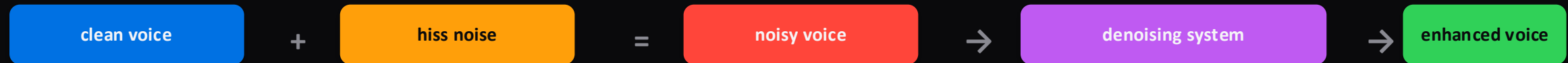
We simulate the signal instead of recording a complex station environment.

This keeps the noise controllable, the experiment repeatable, and the result easier to analyze.

PLACEHOLDER
scenario / setup diagram

Problem statement

Input, output, and design goal



Goal: suppress hiss while preserving useful speech information.

This is a tradeoff, not a perfect-recovery problem.

Signal model

A Signals and Systems viewpoint

$$x_{\text{noisy}}[n] = x_{\text{clean}}[n] + n_{\text{hiss}}[n]$$

clean voice and hiss are both signals

the noisy voice is a superposition

the filter is an input-output system

frequency response explains what the system keeps or suppresses

PLACEHOLDER
input-output system diagram

Visualizing the signals

Waveform, spectrum, and spectrogram

waveform
placeholder

FFT spectrum
placeholder

spectrogram
placeholder

Waveform shows time-domain shape. FFT shows overall frequency energy. Spectrogram shows how energy changes over time and frequency.

Frequency-domain observation

Why hiss suggests a filtering strategy

Clean voice has structured low-to-mid frequency components.

Hiss noise raises the high-frequency background.

The noisy signal makes the high-frequency region visibly stronger.

PLACEHOLDER
spectrum / spectrogram evidence

Baseline: low-pass filter

A fixed frequency-domain mask

Low-pass filtering keeps lower frequencies and attenuates higher frequencies.

It is easy to explain and closely related to frequency response.

In mask language, low-pass is a fixed mask: pass below cutoff, suppress above cutoff.

PLACEHOLDER
low-pass response / fixed mask

Baseline parameter choice

Why cutoff is a design decision

Lower cutoff: stronger noise reduction, but speech becomes duller.

Higher cutoff: more speech detail, but more hiss remains.

The chosen cutoff is a balance between suppression and preservation.

PLACEHOLDER
cutoff comparison figure

Why low-pass is not enough

The baseline is explainable, but coarse

Low-pass uses the same attenuation pattern for the entire signal.

It does not know whether a time-frequency region is speech-dominant or noise-dominant.

This can reduce hiss, but it can also remove useful high-frequency speech detail.

This motivates a more flexible frequency-domain mask.

Frequency-domain masking

A more flexible version of the same idea

$$Y(f) = M(f)X(f)$$

$$Y(t, f) = M(t, f)X(t, f)$$

M close to 1: preserve this component

M close to 0: suppress this component

Low-pass is a fixed mask; frequency masking can be more adaptive

PLACEHOLDER
mask heatmap

Wiener-style mask

A principled way to design the mask

$$M(f) = P_{\text{speech}}(f) / (P_{\text{speech}}(f) + P_{\text{noise}}(f))$$

If speech power is stronger, the mask moves closer to 1.

If noise power is stronger, the mask moves closer to 0.

This follows the intuition of Wiener filtering, without claiming an industrial-grade denoising system.

PLACEHOLDER
Wiener gain curve / mask

Improved method workflow

From STFT to masked enhanced voice



The improved method still uses the frequency-domain story, but makes the attenuation pattern more selective.

Mask visualization

The main new figure for the upgraded demo

Noisy spectrogram
placeholder

Wiener-style mask
placeholder

Masked spectrogram
placeholder

A good result page should show the logic visually: Noisy -> Mask -> Enhanced.

Result comparison

Baseline versus Wiener-style masking

Noisy voice
spectrogram / audio label

Low-pass result
spectrogram / audio label

Mask result
spectrogram / audio label

The goal is not to crown a perfect method, but to explain how a more flexible mask changes the filtering behavior.

Demo

Audio slots to fill after the mask experiment is generated

AUDIO SLOT
clean voice

AUDIO SLOT
noisy voice

AUDIO SLOT
low-pass result

AUDIO SLOT
mask result

Play the same speech segment for every version.

Then connect what we hear to the spectrogram and mask figures.

Keep the demo short enough that the audience remembers the contrast.

Limitations

What this system still cannot solve

The noise is simulated and simpler than a real station environment.

Mask quality depends on how well the noise power is estimated.

If speech and noise overlap strongly in the same time-frequency region, separation becomes difficult.

More advanced methods may require adaptive filtering or learning-based denoising.

PLACEHOLDER
limitations / extension
diagram

A simple filter can become a richer system.

Conclusion

We modeled voice denoising as a Signals and Systems problem.

Low-pass filtering provides a clear baseline.

Frequency-domain masking extends the same idea in a more flexible way.

A Wiener-style mask gives a principled and visual way to design frequency-dependent attenuation.